

MEASUREMENT SCALES

Measurements figures permit statistical analysis and communication of measurement results.

Scaling is an extension of measurement. Scaling involves creating a continuum on which measurements are made.

Types of Measurement Scales

There are four types of measurement scales

1. Nominal Scale,
2. Ordinal Scale,
3. Interval Scale & 4. Ratio Scale.

The choice of the measurement scale has implications for the statistical technique to be used for data analysis.

Nominal Scale: This is the lowest level of measurement. Here, numbers are assigned for the purpose of identification of the objects. Any object which is assigned a higher number is in no way superior to the one which is assigned a lower number. The objects are divided into mutually exclusive and collectively exhaustive categories. Each number is assigned to only one object and each object has only one number assigned to it.

Example:

 What is your religion?

1. Hinduism
2. Sikhism
3. Christianity
4. Islam
5. Any other, (please specify)

Any religion which is assigned a higher number is in no way superior or inferior to the one which is assigned a lower number. The assignment of numbers is only for the purpose of identification.

Nominal scale measurements are used for identifying attributes, such as food habits (vegetarian or non-vegetarian), gender (male/female), caste, respondents, marital status, brands, etc.

The assigned numbers cannot be added, subtracted, multiplied or divided.

The only arithmetic operations that can be carried out are the count of each category.

Therefore, a frequency distribution table can be prepared for the nominal scale variables and mode of the distribution can be worked out.

Ordinal scale: This is the next higher level of measurement than the nominal scale measurement.

An ordinal scale measurement tells whether an object has more or less of characteristics than some other objects. However, it cannot answer how much more or how much less.

Example:

 Rank the following attributes while choosing a restaurant for dinner. The most important rank one, the next important rank of 2 and so on.

Food quality [] Prices [] Menu variety []
Ambience [] Service []

In the ordinal scale, the assigned ranks cannot be added, multiplied, subtracted or divided. One can compute median, percentiles and quartiles of the distribution. The other major statistical analysis which can be carried out is the rank order correlation coefficient. All the statistical techniques which are applicable in the case of nominal scale measurement can also be used for the ordinal scale measurement.

Interval scale: The interval scale measurement is the next higher level of measurement. It takes care of the limitation of the ordinal scale measurement where the difference between the score on the ordinal scale does not have any meaningful interpretation. In the interval scale the difference of the score on the scale has meaningful interpretation. It is assumed that the respondent is able to answer the questions on a continuum scale. The mathematical form of the data on the interval scale may be written as

$$Y = a + b X \quad \text{where } a \neq 0$$

In the interval scale, the difference in score has a meaningful interpretation while the ratio of the score on this scale does not have a meaningful interpretation.

This can be seen from the following interval scale question:

 How likely are you to buy a new designer carpet in the next six months?

.	Very unlikely	Unlikely	Neutral	Likely	Very likely
Scale A	1	2	3	4	5
Scale B	0	1	2	3	4
Scale C	-2	-1	0	1	2

Suppose a respondent ticks the response category 'likely' and another respondent ticks the category 'unlikely'. If we use any of the scales A, B or C, we

note that the difference between the scores in each case is 2. Whereas, when the ratio of the scores is taken, it is 2, 3 and -1 for the scales A, B and C respectively.

Therefore, the ratio of the scores on the scale does not have a meaningful interpretation. The following are some examples of interval scale data.

 How important is price to you while buying a car?

Least important, Unimportant, Neutral, Important, Most important (1 2 3 4 5)

 How do you rate the work environment of your organization?

Very good, Good, Neither good nor bad, Bad, Very bad (5 4 3 2 1)

 How expensive is the restaurant 'XYZ'?

Extremely expensive, Definitely expensive, Somewhat expensive, Somewhat inexpensive, Definitely inexpensive, Extremely inexpensive (1 2 3 4 5 6)

The numbers on this scale can be added, subtracted, multiplied or divided. One can compute arithmetic mean, standard deviation, correlation coefficient, regression analysis and factor analysis. All the techniques applicable for the ordinal and the nominal scale data can also be used for interval scale data.

Ratio scale: This is the highest level of measurement and takes care of the limitations of the interval scale measurement.

The ratio scale measurement can be converted into interval, ordinal and nominal scale. But the other way round is not possible. The mathematical form of the ratio scale data is given by

$$Y = b X.$$

Examples of the ratio scale data are weight, distance travelled, income and sales of a company, to mention a few.

CLASSIFICATION OF SCALES

One of the ways of classifications of scales is in terms of the number of items in the scale. Based upon this, the following classification may be proposed:

I. Single Item vs Multiple Item Scale

Single item scale: In the single item scale, there is only one item to measure a given construct. Example:

Consider the following question:

 How satisfied are you with your current job?

Very Dissatisfied, Dissatisfied, Neutral, Satisfied, Very satisfied

The problem with the above question is that there are many aspects to a job, like pay, work environment, rules and regulations, security of job and relationship with the seniors. The respondent may be satisfied on some of the factors but may not on others. By asking a question as stated above, it will be difficult to analyse the problem areas. To overcome this problem, a multiple item scale is proposed.

Multiple item scale: In multiple item scale, there are many items (factors) that affect the construct that the researcher is trying to measure. Each of the item forms some part of the construct (satisfaction) which the researcher is trying to measure. Example

■ How satisfied are you with the pay you are getting on your current job?

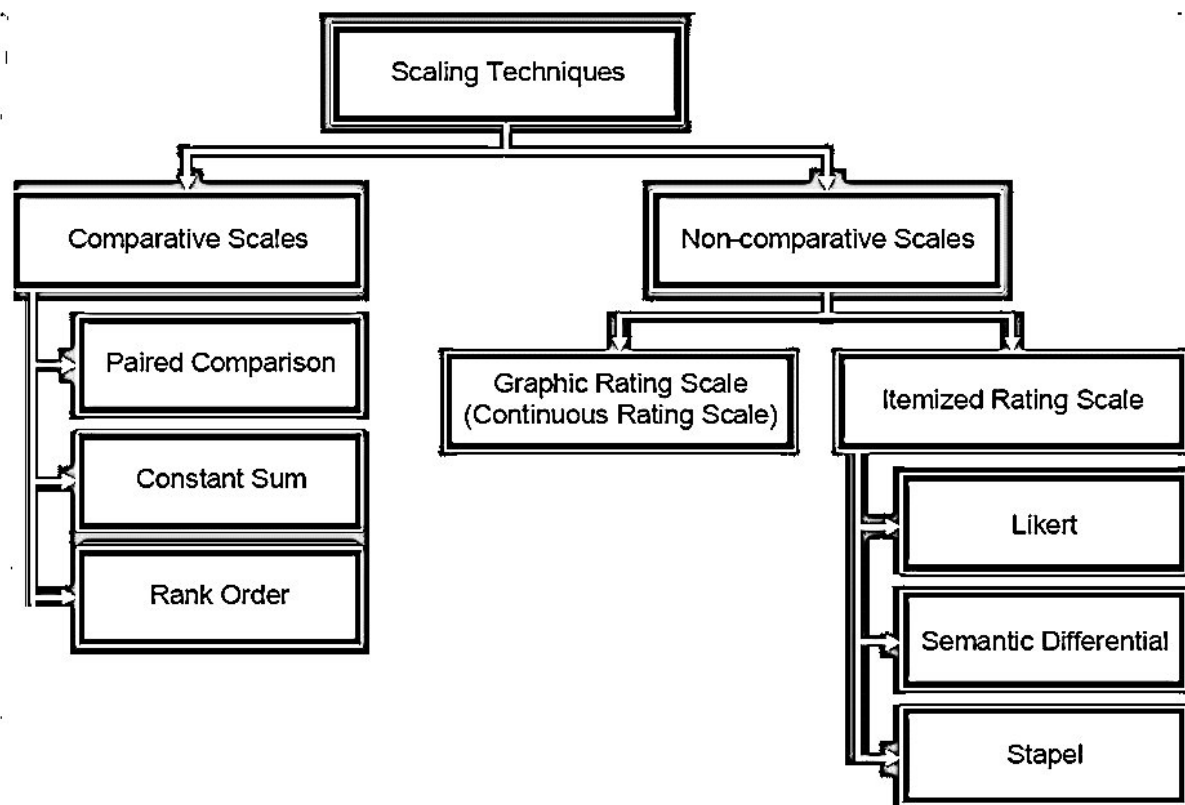
Very dissatisfied, Dissatisfied, Neutral, Satisfied, Very satisfied

■ How satisfied are you with the rules and regulations of your organization?

Very dissatisfied, Dissatisfied, Neutral, Satisfied, Very satisfied

II. Comparative vs Non-comparative Scales

The scaling techniques used in research can also be classified into comparative and non-comparative scales.



i) Comparative scales

In comparative scales it is assumed that respondents make use of a standard frame of reference before answering the question. For example:

Paired comparison scales: Here a respondent is presented with two objects and is asked to select one according to whatever criterion he or she wants to use.

Example:

 Please rate Domino's in comparison to Pizza Hut on the basis of your satisfaction level on the following parameters:

(1 = Much Worse, 2 = Worse, 3 = Same, 4 = Better, 5 = Much Better.)

- a. Variety of menu options [] b. Value for money []
c. Speed of service (delivery time) [] d. Food quality []

The resulting data from this scale is ordinal in nature.

Rank Order Scale: In the rank order scaling, respondents are presented with several objects simultaneously and asked to order or rank them according to some criterion. Consider, for example the following question:

 Rank the following soft drinks in order of your preference, the most preferred soft drink should be ranked one, the second most preferred should be ranked two, and so on.

Coke [] Pepsi [] Limca [] Sprite []
Mirinda [] Fanta [] Seven Up []

Like paired comparison, this approach is also comparative in nature. The problem with this scale is that if a respondent does not like any of the above mentioned soft drink and is forced to rank them in the order of his choice.

The rank order scaling results in the ordinal data.

Constant Sum Rating Scale: In constant sum rating scale, the respondents are asked to allocate a total of 100 points between various objects and brands. The respondent distributes the points to the various objects in the order of his preference. Example:

 Allocate a total of 100 points among the various schools into which you would like to admit your child. The points should be allocated in such a way that the sum total of the points allocated to various schools adds up to 100.

DPS	Mother's Integral
Sai International	DAV Public School

This type of data is not only comparative in nature but could also result in ratio scale measurement.

ii) Non-comparative scales

In the non-comparative scales, the respondents do not make use of any frame of reference before answering the questions. The resulting data is generally assumed to be interval or ratio scale.

The non-comparative scales are divided into two categories, namely, the Graphic Rating scales and the Itemized rating scales.

Graphic rating scale: This is a continuous scale, also called graphic rating Scale. In the graphic rating scale the respondent is asked to tick his preference on a graph. Example:

 Please put a tick mark (📍) on the following line to indicate your preference for fast food.



To measure one has to measure the distance from the extreme left to the position where a tick mark has been put.

Itemized rating scale

In the itemized rating scale, the respondents are provided with a scale that has a number of brief descriptions associated with each of the response categories. There are certain issues that should be kept in mind while designing the itemized rating scale. These issues are:

Number of categories to be used: There is no hard and fast rule as to how many categories should be used in an itemized rating scale. However, it is a practice to use five or six categories. Some researchers are of the opinion that more than five categories should be used in situations where small changes in attitudes are to be measured. There are others that argue that the respondents would find it difficult to distinguish between more than five categories.

Odd or even number of categories: It has been a matter of debate among the researchers as to whether odd or even number of categories are to be used. By using even number of categories the scale would not have a neutral category and the respondent will be forced to choose either the positive or the negative side of the attitude. If odd numbers of categories are used, the respondent has the freedom to be neutral if he wants to be so.

Balanced versus unbalanced scales: A balanced scale is the one which has equal number of favourable and unfavourable categories. The following is the example of a balanced scale:

🎬 How important is price to you in buying a new car?

Very important, Relatively important, Neither important nor unimportant, Relatively unimportant, Very unimportant

In this question, there are five response categories, two of which emphasize the importance of price and two others that do not show its importance. The middle category is neutral.

The following is the example of the unbalanced scale.

🎬 How important is price to you in buying a new car?

Extremely important, Important, Somewhat important, Unimportant
In this question there are four response categories that are skewed towards the importance given to the price, whereas one category is for the unimportant side. Therefore, this question is an unbalanced question.

Forced versus non-forced scales: In the forced scale, the respondent is forced to take a stand, whereas in the non-forced scale, the respondent can be neutral if he/she so desires. The argument for a forced scale is that those who are reluctant to reveal their attitude are encouraged to do so with the forced scale. Paired comparison scale, rank order scale and constant sum rating scales are examples of forced scales.

Likert scale: This is a multiple item agree-disagree five-point scale. The respondents are given a certain number of items (statements) on which they are asked to express their degree of agreement/disagreement. This is also called a summated scale because the scores on individual items can be added together to produce a total score for the respondent. An assumption of the Likert scale is that each of the items (statements) measures some aspect of a single common factor, otherwise the scores on the items cannot legitimately be summed up. In a typical research study, there are generally 25–30 items on a Likert scale.

To construct a Likert scale to measure a particular construct, a large number of statements pertaining to the construct are listed. These statements could range from 80–120. The identification of the statements is done through exploratory research which is carried out by conducting a focus group, unstructured interviews with knowledgeable people, literature survey, analysis of case studies and so on.

Suppose we want to assess the image of a company. As a first step, an exploratory research may be conducted by having an informal interview with

the customers, and employees of the company. The general public may also be contacted.

Suppose the number of statements to measure the constructs is 100 in number. Now samples of representative respondents are asked to state their degree of agreement/disagreement on those statements.

Table gives a few statements to assess the image of the company.

Likert Scale Statements to Measure the Image of the Company

Strongly Disagree (SD) Disagree (D)

Neither Agree nor Disagree (N) Agree (A) Strongly Agree (SA)

1. The company makes quality products ____
2. It is a leader in technology ____
3. It doesn't care about the general public. _____
4. The company leads in R&D to improve products. _____
5. The company is not a good paymaster. ____
6. The products of the company go through stringent quality tests. ____
7. The company has not done anything to curb pollution. ____
8. It does not care about the community near its plant. ____
9. The company's stocks are good to buy or own. ____
10. The company does not have good labour relations. _____

The scale contains statements' some of which are favourable to the construct we are trying to measure and some are unfavourable to it.

Example, statements numbering 1, 2, 4, 6 and 9 are favourable statements, whereas the remaining are unfavourable statements. Generally, in a Likert scale, there is an approximately equal number of favourable and unfavourable statements. After the scale is administered, numerical values are assigned to the responses. The rule is that a 'strongly agree' response for a favourable statement should get the same numerical value as the 'strongly disagree' response of the unfavourable statement.

What is the reason for having a mixture of favourable and unfavourable statements in a Likert scale?

- The responses by the respondent should not become monotonous.
- It serves Validity and Reliability

For 10 statements maximum score is 50 and minimum is 10.

Now if there are 100 statements, the score on the image of the company can be worked out for each respondent by adding his/her scores on the 100

statements. The minimum score for each respondent will be 100, whereas the maximum score would be 500.

A typical Likert scale comprises about 25–30 statements. In order to select 25 statements from the 100 statements, we need to discard some of them. The rule behind discarding the statements is that those items that are non-discriminating should be removed.

MEASUREMENT ERROR

Measurement error occurs when the observed measurement on a construct or concept deviates from its true values. The following is a list of the sources of measurement errors.

■ There are factors like mood, fatigue and health of the respondent which may influence the observed response while the instrument is being administered. The other factors could be education, job, awareness of topic and reluctance to express an opinion.

■ At times, the errors may be committed at the time of coding, entering of data from questionnaire to the spreadsheet on the computer and at the tabulation stage. The other reasons could be defective instrument for data collection like lengthy and ambiguous questionnaire with leading questions (suggestive responses) in the instrument.

The observed measurement in any research need not be equal to the true measurement. The observed measurement can be written as

$$O = T + S + R \quad \text{Where,}$$

O = Observed measurement

T = True score

S = Systematic error

R = Random error

It may be noted that the total error consists of two components—systematic error and random error. Systematic error causes a constant bias in the measurement.

Suppose there is a weighing scale that weighs 50 gm less for every one kg of product being weighed. The error would consistently remain the same irrespective of the kind of product and the time at which product is weighed. Random error on the other hand involves influences that bias the measurements but are not systematic.

Suppose we use different weighing scales to weigh one kg of a product and if systematic error is assumed to be absent, we may find that recorded weights

may fall within a range around the true value of the weight, thereby causing random error.

Criteria for Good Measurement

There are three criteria for evaluating measurements:

(1) Reliability, (2) Validity and (3) Sensitivity.

1. Reliability

Test-retest reliability: In this method, repeated measurements of the same person or group using the same scale under similar conditions are taken. A very high correlation between the two scores indicates that the scale is reliable.

Split-half reliability method: This method is used in the case of multiple item scales. Here the number of items is randomly divided into two parts and a correlation coefficient between the two is obtained. A high correlation indicates that the internal consistency of the construct leads to greater reliability.

2. Validity

The validity of a scale refers to the question whether we are measuring what we want to measure. Validity of the scale refers to the extent to which the measurement process is free from both systematic and random errors.

Content validity: This is also called face validity. It involves subjective judgement by an expert for assessing the appropriateness of the construct. For example, to measure the perception of a customer towards an Airlines, a multiple item scale is developed. However researcher might have missed some important item.

In order to judge the content validity of these items, a set of experts may be requested to examine the representativeness of the 15 items.

Predictive validity: This involves the ability of a measured phenomena at one point of time to predict another phenomenon at a future point of time. If the correlation coefficient between the two is high, the initial measure is said to have a high predictive ability. E.g. CAT and MAT score, UPSC and OPSC

3. Sensitivity

Sensitivity refers to an instrument's ability to accurately measure the variability in a concept. A dichotomous response category such as agree or disagree does not allow the recording of any attitude changes. A more sensitive measure with numerous categories on the scale may be required.

than a single-item scale.

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7. What are systematic errors?

8. Name the method in which the number of items is randomly divided into two parts and a correlation coefficient between the two is obtained.

5.5 ANSWERS TO CHECK YOUR PROGRESS

QUESTIONS

1. One of the limitation of the nominal scale measurements is that we cannot say whether the assigned number to an object is higher or lower than the one assigned to another option. The ordinal scale measurement takes care of this limitation.

2. The highest level of measurement is the ratio scale.
3. The intention component of attitude also called the behavioural component, reflects a predisposition to an action by reflecting the consumer's buying or purchase intention.
4. The Q-sort technique makes use of the rank order procedure in which objects are sorted into different piles based on their similarity with respect to certain criterion.
5. Some examples of forced scales include paired comparison scale, rank order scale and constant sum rating scales.
6. The Likert scale is also called as summated scale because the scores on individual items can be added together to produce a total score for the respondent.
7. Systematic errors are one of the components of total error which causes a constant bias in the measurement.
8. Split-half reliability method is the method in which the number of items is randomly divided into two parts and a correlation between the two is obtained.

5.6 SUMMARY

 Measurement means the assignment of numbers or other symbols to the characteristics of certain objects. Scaling is an extension of measurement. Scaling involves creating a continuum on which measurements on the objects are located. There are four types of measurement scales: nominal, ordinal, interval and ratio scale.

Attitude Measurement
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 Attitude is a predisposition of the individual to evaluate some objects or symbol. Attitude has three components: cognitive, affective and intention or action component.

 Scales can be classified as single-item and multiple-item scales. Another classification could be whether the scales are comparative or noncomparative

in nature.

■ The observed measurement need not be equal to the true value of the measurement. Some systematic and random errors may be found in the observed measurement. There are three criteria for determining the accuracy of a measurement—reliability, validity and sensitivity.

5.7 KEY WORDS

■ **Balanced scale:** A scale that has equal number of favourable and unfavourable categories.

■ **Comparative scale:** A scale in which respondents make use of some standard frame of reference before answering the question.

■ **Forced scale:** A scale in which the respondent is forced to take a stand

■ **Interval scale:** A scale that makes use of an arbitrary origin.

■ **Validity:** It deals with whether a scale measures what it is supposed to measure.

5.8 SELF ASSESSMENT QUESTIONS AND EXERCISES

Short-Answer Questions

1. What is the meaning of measurement in research? Give examples.
2. Outline the steps involved in constructing a Likert scale.
3. Briefly explain the concepts of reliability, validity and sensitivity.

Long-Answer Questions

1. Discuss four types of measurements using examples.
2. Define attitude. What are its various components?
3. Explain an itemized rating scale. What are the various issues involved in constructing an itemized rating scale?

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Attitude Measurement